QUALIFICATION REQUIREMENTS REMOVED

MIL-M-38510/472A(NASA)
9 August 1983
SUPERSEDING
MIL-M-38510/472(NASA)
20 June 1980

MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, CMOS,
512 WORD x 8-BIT STATIC READ ONLY MEMORY (ROM)
MONOLITHIC SILICON

INACTIVE FOR NEW DESIGN AFTER DATE OF THIS REVISION

This specification is approved for use by George C. Marshall Space Flight Center, National Aeronautics and Space Administration, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, silicon gate CMOS microcircuit. One product assurance class and a choice of case outlines and lead finishes are provided for each type and are reflected in the complete part number.

1.2 Part number. The part number shall be as specified in MIL-M-38510, except the "JAN" or "J" certification shall not be used.

1.2.1 Device type. The device type shall be as follows:

Device type	Circuit
01	512 Word x 8-Bit Static ROM

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outline. The case outline shall be designated as follows:

Outline letter	Case outline (see MIL-M-38510, appendix C)
J	D-3 (24-lead, 1/2" x 1 1/4"), dual-in-line package.
K	F-6 (24-lead, 3/8" x 5/8"), flat package.

1.3 Absolute maximum ratings.

Storage temperature range (T_{stg})	-65°C to +150°C
Case operating temperature range (T_C)	-55°C to +125°C
Supply voltage range (V_{DD} - V_{SS})	-0.5 V dc to +13 V dc
Power dissipation per package (P_D):	
For T_C = -55°C to +100°C	500 mW
For T_C = +100°C to +125°C	Derate linearly at 12 mW/°C to 200 mW
Device dissipation per output transistor:	
For T_C = -55°C to +125°C	100 mW
Input voltage range, all inputs	-0.5 to V_{DD} +0.5 V dc
DC input current, any one input	±10 mA
Lead temperature (soldering, 10 seconds):	
At distance 1/16 ±1/32 inch (1.59 +0.79 mm) from case	+265°C

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: George C. Marshall Space Flight Center, National Aeronautics and Space Administration, ATTN: EGO2, Marshall Space Flight Center, AL 35812, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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Maximum junction temperature (T_J)	+150°C
Thermal resistance, junction to case (θ_{JC}):	
Case J	0.0833°C/mW
Case K	0.15°C/W

1.4 Recommended operating conditions. (At T_C = -55°C to +125°C.)

Supply voltage range (V_{DD}-V_{SS})	+4.5 V dc minimum to +11 V dc maximum
Input low (V_{IL}) voltage range	0 to +1.5 V dc at V_{DD} = +5 V dc; 0 to +3 V dc at V_{DD} = +10 V dc
Input high (V_{IH}) voltage range	+3.5 V dc to +5 V dc at V_{DD} = +5 V dc; 7 V dc to 10 V dc at V_{DD} = 10 V dc

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications standards, and handbooks, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of specifications, standards, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein. An altered item drawing shall be prepared by the contracting activity to specify the required mask-program.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510, and herein.

3.2.1 Terminal connections. The terminal connections shall be as specified on figure 1.

3.2.2 Functional diagram. The circuit functional diagram shall be as specified on figure 2.

3.2.3 Logic diagram. The logic diagram shall be as specified on figure 3.

3.2.4 Case outlines. The case outlines shall be as specified in MIL-M-38510 and in 1.2.3 herein.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-38510 (see 6.6).

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TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions V_{SS} = 0 V, -55°C ≤ T_A ≤ 125°C Unless otherwise specified	Limits Min	Limits Max	Unit
Positive clamping input to V_{DD}	$V_{IC}(pos)$	T_C = +25°C, V_{DD} = 0 V, V_{SS} open, outputs open, I_I = 1 mA	0.4	1.0	V
Negative clamping input to V_{SS}	$V_{IC}(neg)$	T_C = +25°C, V_{DD} open, V_{SS} = 0 V, outputs open, I_I = -1 mA	-0.4	-1.0	V
Quiescent supply current	I_{SS}	V_{DD} = +13 V dc, any combination of inputs		-1	mA
High level output current	I_{OH1}	V_{DD} = +10 V dc (see table III for V_I and V_{OH})	-0.95		mA
	I_{OH2}	V_{DD} = +5 V dc (see table III for V_I and V_{OH})	-0.45		mA
Low level output current	I_{OL1}	V_{DD} = +10 V dc (see table III for V_I and I_{OL})	0.95		mA
	I_{OL2}	V_{DD} = +5 V dc (see table III for V_I and I_{OL})	0.45		mA
High impedance 3-state output leakage current	I_Z	V_{DD} = +13 V (see table III for V_I)		±300	nA
Input leakage current (any input)	I_{IH}	V_{DD} = +13 V, V_I = +13 V		300	nA
	I_{IL}	V_{DD} = +13 V, V_I = 0 V		-300	nA
Break-down supply current	I_{BKDN}	V_{DD} = +13 V (see table III for V_I)		10	mA
Address access time	t_{AA}	V_{DD} = +5 V, C_L = 50 pF, R_L = 200 kΩ V_{DD} = +10 V, C_L = 50 pF, R_L = 200 kΩ		1300 550	ns
Access time from chip select	t_{ACS}	V_{DD} = +5 V, C_L = 50 pF, R_L = 200 kΩ V_{DD} = +10 V, C_L = 50 pF, R_L = 200 kΩ		1200 550	ns
Chip select delay time from chip select to data valid	t_{CS}	V_{DD} = +5 V, C_L = 50 pF, R_L = 200 kΩ V_{DD} = +10 V, C_L = 50 pF, R_L = 200 kΩ		780 390	ns
Time from chip select to high impedance output	t_{CSZ}	V_{DD} = +5 V, C_L = 50 pF, R_L = 200 kΩ V_{DD} = +10 V, C_L = 50 pF, R_L = 200 kΩ		780 390	ns

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3.4 Electrical performance characteristics. The electrical performance characteristics are as specified in table I, and apply over the full recommended case operating temperature range, unless otherwise specified.

3.5 User mask-program. Since the ROM is memory programmed by the manufacturer in a wide variety of configurations, the contracting activity shall provide an altered item drawing describing the mask-program to be used by the manufacturer. All testing requirements and quality assurance provisions herein, including the requirements of the altered item drawing, shall be satisfied by the manufacturer prior to delivery.

TABLE II. Electrical test requirements.

MIL-STD-883 test requirements	Subgroups (see table III unless otherwise specified)
	Class B devices
Interim electrical parameters (pre burn-in) (method 5004)	1
Interim electrical parameters (between static burn-in and dynamic burn-in) (method 5004)	Not applicable
Final electrical test parameters (method 5004)	1*,2,7*,8
Group A test requirements (method 5005)	1,2,7,8,9
Group B electrical parameters (method 5005)	None
Group B V_{ZAP} (method 5005)	See 4.5.3
Group C end-point electrical parameters (method 5005)	1,2,7,8
Additional electrical subgroups for group C periodic inspections (method 5005)	10
Group D end-point electrical parameters (method 5005)	1,2,7,8

*PDA applies to subgroup 1 and 7 (see 4.2b).

3.6 Electrical test requirements. The electrical test requirements shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III.

3.7 Marking. Marking shall be in accordance with MIL-M-38510 and 1.2 herein. The altered item drawing number describing the user mask-program shall be added to the marking by the manufacturer. At the option of the manufacturer, marking of the country of origin may be omitted from the body of the microcircuit, but shall be retained on the initial container. The "JAN" or "J" certification mark shall not be used.

3.8 Manufacturer eligibility. To be eligible to supply microcircuits to this specification, a manufacturer shall have a manufacturer certification in accordance with MIL-M-38510 for at least one line, not necessarily the line producing the device type described herein.

3.9 Certification. Certification in accordance with MIL-M-38510 is not required for this device.

4. QUALITY ASSURANCE PROVISIONS

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4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007, as applicable, of MIL-STD-883, except as modified herein.

4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to quality conformance inspection. The following additional criteria shall apply:

a. Burn-in test (method 1015 of MIL-STD-883).
   (1) The burn-in test as described in 3.1.10 of method 5004 of MIL-STD-883 shall be performed as indicated. (3.1.11 and 3.1.12 are not required; 3.1.9 is required for class B devices.)
   (2) For class B devices, dynamic burn-in shall be performed for 160 hours minimum at 125°C ambient temperature, using the circuit shown on figure 4, or equivalent.
b. The percent defective allowable (PDA) for class B devices shall be 10 percent based on failures from group A, subgroups 1 and 7 tests after cooldown as final electrical test in accordance with method 5004 of MIL-STD-883, and with no intervening electrical measurements. If interim electrical parameter tests are performed prior to burn-in, failures resulting from pre burn-in screening may be excluded from the PDA. If interim electrical parameter tests prior to burn-in are omitted, then all screening failures shall be included in the PDA. The verified failures of group A, subgroups 1 and 7, after burn-in divided by the total number of devices submitted for burn-in in that lot shall be used to determine the percent defective for that lot, and the lot shall be accepted or rejected based on the PDA for the applicable device class.
c. Interim and final electrical test parameters shall be as specified in table II, except interim electrical parameters test prior to burn-in is optional at the discretion of the manufacturer.
d. All devices shall be mask-programmed to the requirements of the altered item drawing prior to initiation of any testing.

4.3 Qualification inspection. Qualification inspection is not required.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4). Generic test data (see 6.7) may be used to satisfy the requirements for groups C and D inspections. Quality conformance inspection shall be completed on the specific devices covered by this specification before they are shipped.

4.4.1 Group A inspection. Group A inspection shall consist of the test subgroups and LTPD values shown in table I of method 5005 of MIL-STD-883 and as follows:

a. Tests shall be as specified in table II herein.
b. Subgroups 4, 5, and 6 of method 5005 of MIL-STD-883 shall be omitted.

4.4.2 Group B inspection. Group B inspection shall consist of the test subgroups and LTPD values shown in table II of method 5005 of MIL-STD-883. A special subgroup shall be added to the group B inspection requirements for class B devices and shall consist of the test, conditions and limits specified in 4.5.2. The LTPD for this subgroup shall be 15.

4.4.3 Group C inspection. Group C inspection shall consist of the test subgroups and LTPD value shown in tables III and IV of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical test parameters shall be as specified in table II herein.
b. Steady-state life test (method 1005 of MIL-STD-883) conditions:
   (1) Test condition D or E, and as shown on figure 4, or equivalent.
   (2) T_A = +125°C minimum.
c. Additional electrical subgroups as specified in table II herein shall be added to the group C inspection requirements for class B device.

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Cases J and K

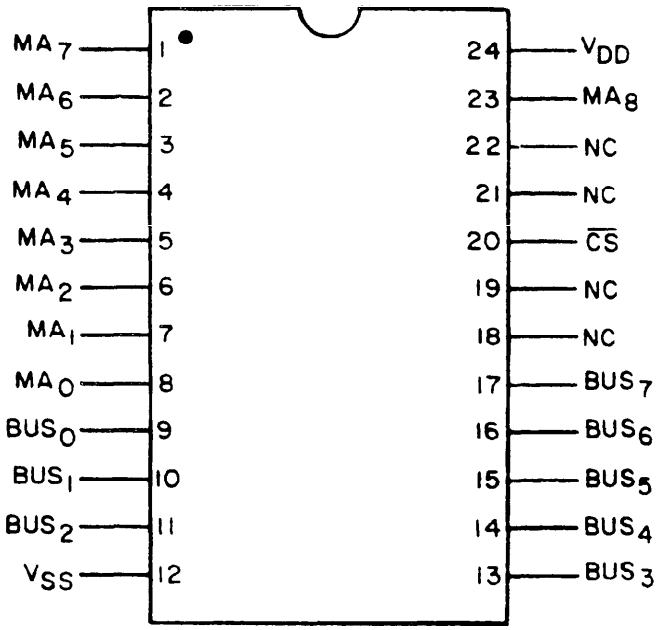


FIGURE 1. Terminal connections.

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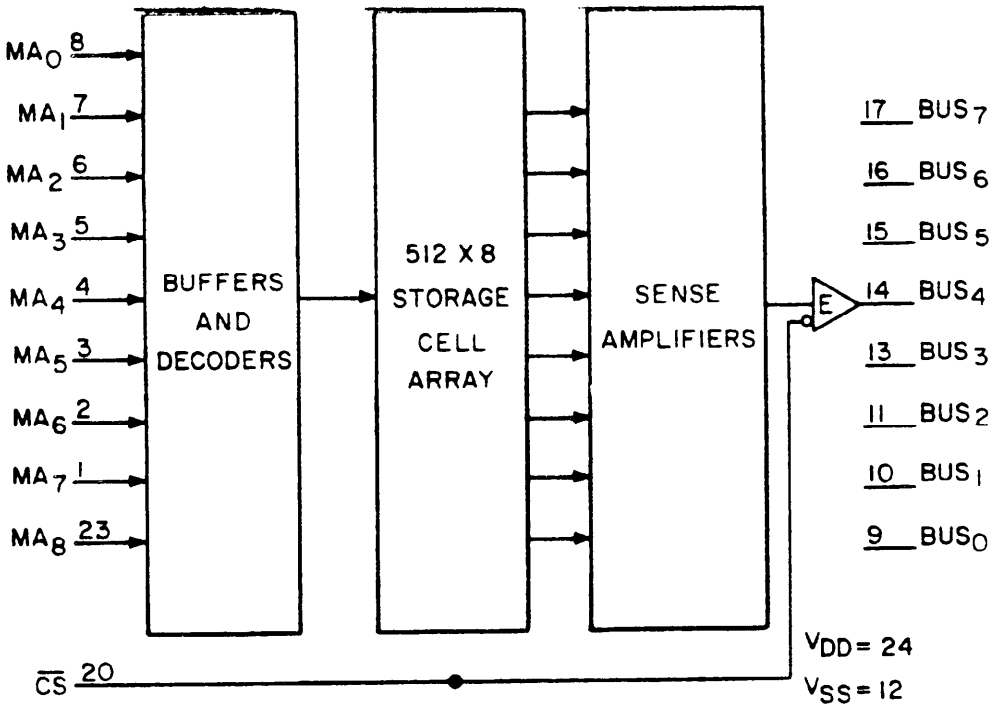


FIGURE 2. Functional diagram.

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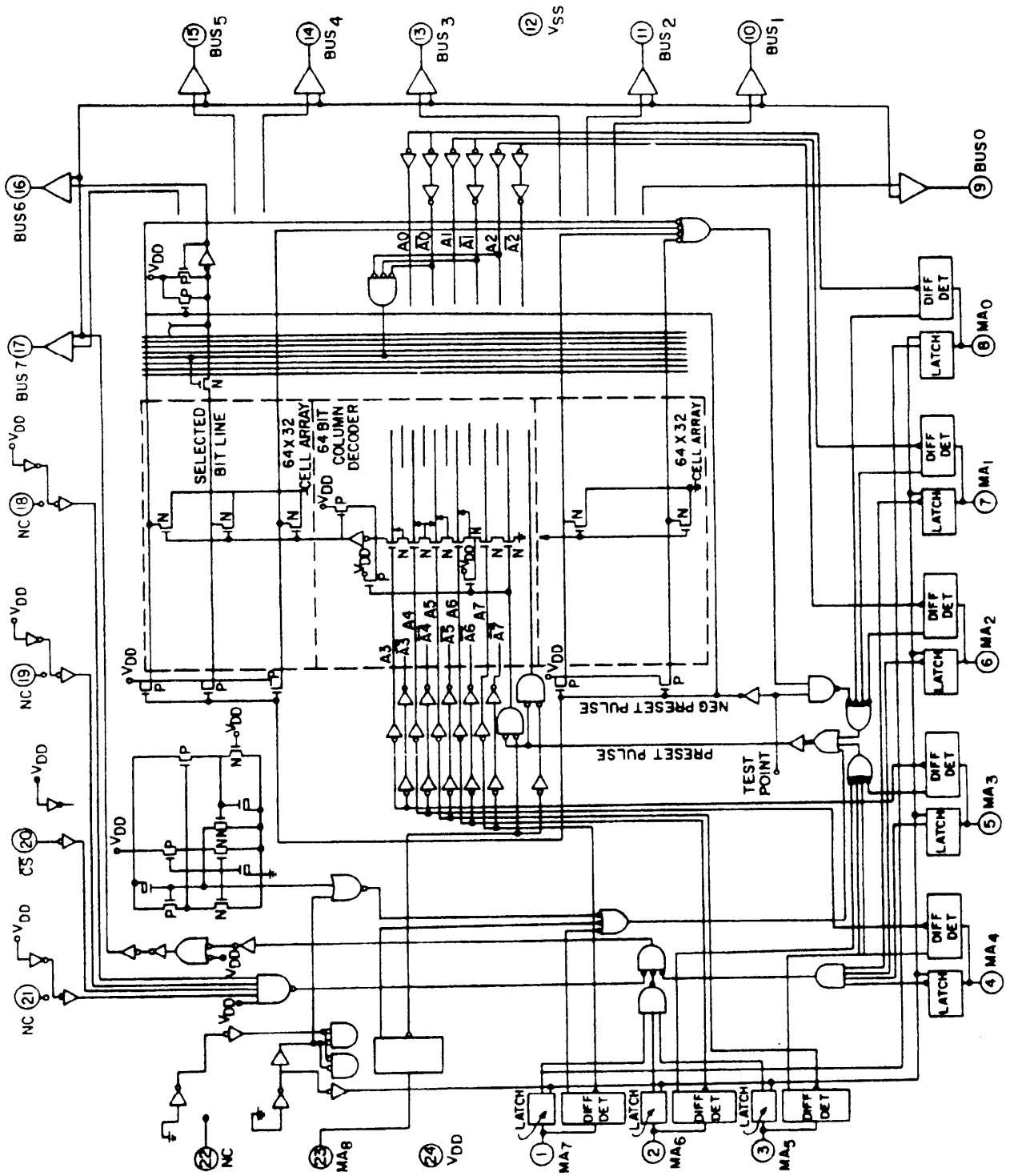
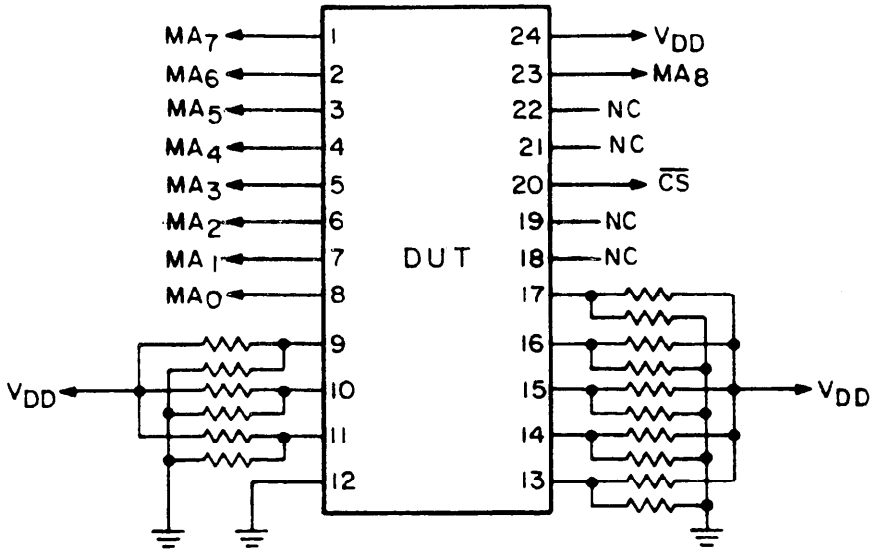
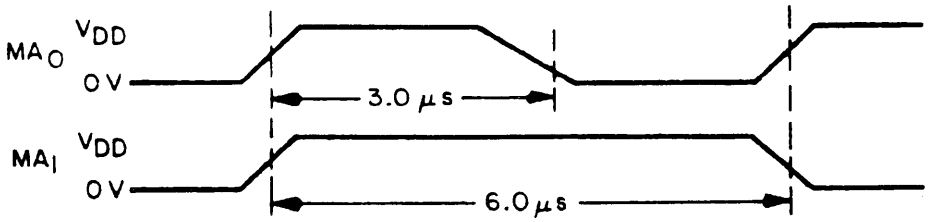


FIGURE 3. Logic diagram.

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NOTES:
1. All resistors shall be 14 kΩ ±5%.
2. MA_0 through MA_8 are division by 2 based on MA_0.
3. V_{DD} = 13 V.

FIGURE 4. Dynamic burn-in and steady state life test circuit.

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TABLE III. Group A inspection for device type 01.

Symbol	MIL-STD-883 method	Cases X, Y	Terminal conditions and limits (pins not designated are open)																								Measured terminal	Test limits						Units
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24		Subgroup 1 T_C = 25°C		Subgroup 2 T_C = 125°C		Subgroup 3 T_C = -55°C		
		Test No.	MA_7	MA_6	MA_5	MA_4	MA_3	MA_2	MA_1	MA_0	BUS_0	BUS_1	BUS_2	V_{SS}	BUS_3	BUS_4	BUS_5	BUS_6	BUS_7	NC	NC	$\overline{CS}$	NC	NC	MA_8	V_{DD}		Min	Max	Min	Max	Min	Max	
V_{IC} (pos)		1	1mA																							GND	1	0.4	1.0					V
		2		1mA																							2							
		3			1mA																						3							
		4				1mA																					4							
		5					1mA																				5							
		6						1mA																			6							
		7							1mA																		7							
		8								1mA																	8							
		9																				1mA					20							
		10																							1mA		21							
V_{IC} (neg)		11	-1mA											GND													1	-0.4	-1.0					
		12		-1mA																							2							
		13			-1mA																						3							
		14				-1mA																					4							
		15					-1mA																				5							
		16						-1mA																			6							
		17							-1mA																		7							
		18								-1mA																	8							
		19																				-1mA					20							

See footnotes at end of table III.

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MIL-M-38510/472A (USA)

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TABLE III. Group A inspection for device type 01 - Continued

Symbol	MIL-STD-883 method	Cases J, K	Terminal conditions and limits (pins not designated are open)																								Measured terminal	Test limits						Unit
																												Subgroup 1 T_C = 25°C		Subgroup 2 T_C = 125°C		Subgroup 3 T_C = -55°C		
		Test No.	1 MA_7	2 MA_6	3 MA_5	4 MA_4	5 MA_3	6 MA_2	7 MA_1	8 MA_0	9 BUS_0	10 BUS_1	11 BUS_2	12 V_{SS}	13 BUS_3	14 BUS_4	15 BUS_5	16 BUS_6	17 BUS_7	18 NC	19 NC	20 $\overline{CS}$	21 NC	22 NC	23 MA_8	24 V_{DD}		Min	Max	Min	Max	Min	Max	
I_{IC} (open)														GND										-1mA		23	-0.4	-1.0					V
I_{RDN}		21	13V	13V	13V	13V	13V	13V	13V	13V												0V			13V	13V	12		-5.0		10			mA
I_{IL} 1/	3009	22	0V	13V	13V	13V	13V	13V	13V	13V												13V			13V	13V	1		-100		-300		-100	nA
		23	13V	0V	13V	13V	13V																				2							
		24		13V	0V	13V	13V																				3							
		25			13V	0V	13V																				4							
		26				13V	0V																				5							
		27					13V	0V																			6							
		28						13V	0V																		7							
		29						13V	13V	0V																	8							
		30						13V	13V	13V												0V					20							
		31						13V	13V	13V												13V			0V		23							
I_{IH} 1/	3010	32	0V	0V	0V	0V	0V	0V	0V	0V												0V			0V		1		100		300		100	
		33	0V	13V	0V	0V	0V	0V																			2							
		34		0V	13V	0V	0V	0V																			3							
		35		0V	0V	13V	0V	0V																			4							
		36		0V	0V	0V	13V	0V																			5							
		37		0V	0V	0V	0V	13V																			6							
		38		0V	0V	0V	0V	0V	13V																		7							

See footnotes at end of table III.

MIL-M-38510/124A

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TABLE III. Group A inspection for device type 01 - Continued.

Symbol	MIL-STD-883 method	Cases	Terminal conditions and limits (pins not designated are oper)																									Mea ter
		Test No.	1 MA_7	2 MA_6	3 MA_5	4 MA_4	5 MA_3	6 MA_2	7 MA_1	8 MA_0	9 BUS_0	10 BUS_1	11 BUS_2	12 V_{SS}	13 BUS_3	14 BUS_4	15 BUS_5	16 BUS_6	17 BUS_7	18 NC	19 NC	20 $\overline{CS}$	21 NC	22 NC	23 MA_8	24 V_{DD}		
I_{IH} 1/	3014	39	0V	0V	0V	0V	0V	0V	0V	13V				GND								0V			0V	13V		
		40								0V												13V			0V			
		41																				0V			13V			
I_{IL} 2/		42									13V	13V	13V		13V	13V	13V	13V	13V			3V			0V			
		43																										
		44																										
		45																										
		46																										
		47																										
		48																										
		49																										
I_{SS1} 1/	3005	50																				0V			13V			
I_{SS2} 1/	3005	51	13V	13V	13V	13V	13V	13V	13V	13V												13V						
I_{OH1} 3/		52									9.5V	9.5V	9.5V		9.5V	9.5V	9.5V	9.5V	9.5V			10V				10V		
		53																										
		54																										
		55																										
		56																										
		57																										
		58																										
		59																										
I_{OH2} 3/		60									4.6V	4.6V	4.6V									5V				5V		
		61																				5V				5V		
		62																				5V				5V		

See footnotes at end of table III.

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TABLE III. Group A inspection for device type 01 - Continued.

Symbol	MIL-STD-883 method	Cases 1, 2	Terminal conditions and limits (pins not designated are open)																								Measured terminal	Test limits						Units
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24		Subgroup 1 T_C = 25°C		Subgroup 2 T_C = 125°C		Subgroup 3 T_C = -55°C		
		Test No.	MA_7	MA_6	MA_5	MA_4	MA_3	MA_2	MA_1	MA_0	BUS_0	BUS_1	BUS_2	V_{SS}	BUS_3	BUS_4	BUS_5	BUS_6	BUS_7	NC	NC	$\overline{CS}$	NC	NC	MA_8	V_{DD}		Min	Max	Min	Max	Min	Max	
I_{OLK} 4/		63												GND	4.6V							5V				5V	13	-0.60		-0.45		-0.70		mA
		64														4.6V											14							
		65															4.6V										15							
		66																4.6V									16							
		67																	4.6V								17							
I_{OL1} 4/		68									.5V											10V				10V	9	1.20		0.95		1.40		
		69										5V															10							
		70											.5V														11							
		71													.5V												13							
		72														.5V											14							
		73															.5V										15							
		74																5V									16							
		75																	.5V								17							
I_{OL2} 4/		76									.4V											5V				5V	9	0.60		0.45		0.70		
		77										.4V															10							
		78											.4V														11							
		79													.4V												13							
		80														.4V											14							
		81															.4V										15							
		82																.4V									16							
		83																	.4V								17							

See footnotes at end of table III

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TABLE III. Group A inspection for device type 01 - Continued.

Symbol	MIL-STD-883 method	Cases J, K	Terminal conditions and limits (pins not designated are open)																								Measu term
		Test No.	1 MA_7	2 MA_6	3 MA_5	4 MA_4	5 MA_3	6 MA_2	7 MA_1	8 MA_0	9 BUS_0	10 BUS_1	11 BUS_2	12 V_{SS}	13 BUS_3	14 BUS_4	15 BUS_5	16 BUS_6	17 BUS_7	18 NC	19 NC	20 $\overline{CS}$	21 NC	22 NC	23 MA_8	24 V_{DD}	
Func-tional tests	3014	84												GND												4.5V	FU
	3014	85												GND												9V	BU
	3014	86												GND												11V	BU
t_{AA}		87												GND												10V	BUS
t_{AA}		88																								5V	BUS
t_{ACS}		89																								10V	BUS
t_{ACS}		90																								5V	BUS
t_{CS} 8/		91																								10V	BUS
t_{CS} 8/		92																								5V	BUS
t_{CSZ} 8/		93																								10V	BUS
t_{CSZ} 8/		94																								5V	BUS

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NOTES:

1/ Leakage current (I_{SS}, I_{IH}, I_{IL}) subgroup 3, shall not be required for 100% screening.

2/ $\overline{CS}$ is high to drive the output to the high-Z state.

3/ Address shall be selected from user mask-program for high output of the tested pin.

4/ Address shall be selected from user mask-program for low output of the tested pin.

5/ Input and output levels for functional tests are derived from a custon ROM test pattern unique to each ROM variant. The test pattern must include a truth table for the 512 byte data array, a truth table for the 2 combinations of $\overline{CS}$.

6/ For V_{DD} = 10 V, V_{IL} = 3.0 V, V_{IH} = 7.0 V

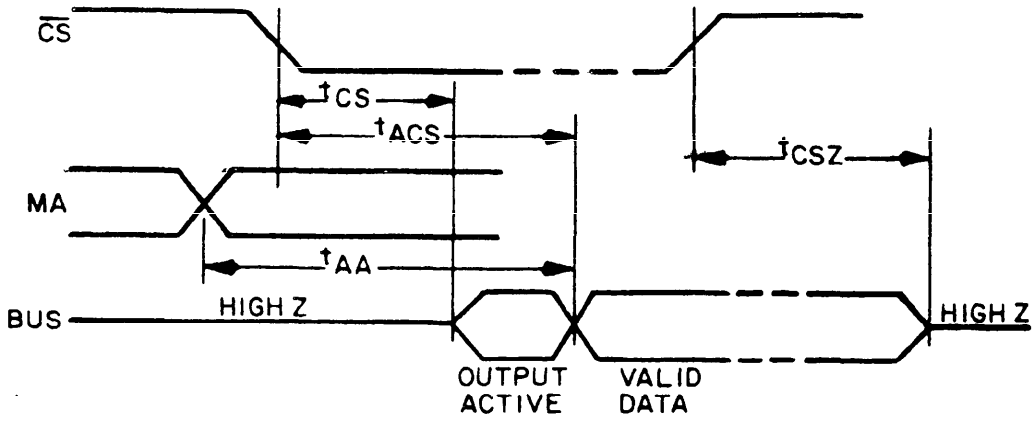
For V_{DD} = 5 V, V_{IL} = 1.5 V, V_{IH} = 3.5 V

V_{OL} < 0.5 V V_{OH} > V_{DD} -0.5 V

7/ GALPAT pattern description

1. Word 0 is read
2. Word 1 is read
3. Word 0 is read
4. Word 2 is read
5. Word 0 is read
6. The reading procedure continues back and forth between the word 0 and the next higher numbered word until word 511 is reached, then increments to the next word and reads back and forth as in steps 1 through 6 until all words are tested.

8/ Timing diagram



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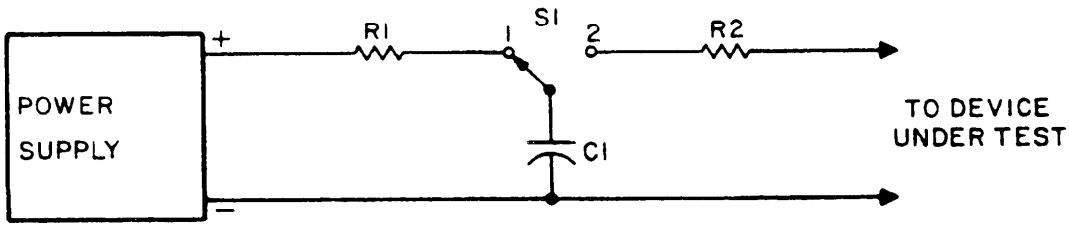
4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883. End-point electrical parameters shall be as specified in table II herein.

4.5 Methods of inspection. Methods of inspection shall be specified as follows.

4.5.1 Voltage and current. All voltages given are referenced to the microcircuit V_{SS} terminal. Currents given are conventional and positive when flowing into the referenced terminal.

4.5.2 High voltage (V_{ZAP}) test of input protection circuits. Unless otherwise specified, all input terminals (up to a maximum of four) of the device under test (DUT) shall be subjected to a voltage pulse from a 100 pF source charged to 400 V (see figure 5). The destructive test shall be conducted in the following sequence:

a. Measure I_{IH} and I_{IL} at inputs selected, as stated above, at 25°C and V_{DD} = 13 V. Also measure I_{SS} at 25°C and V_{DD} = 13 V. These measurements shall be made in accordance with table III herein. The test limits for a single terminal measurement of I_{IH} and I_{IL} shall be 100 nA maximum.



R1 = appropriate current limiting resistance. 10 MΩ ≤ R1 ≤ 50 MΩ.
R2 = 1.5 kΩ.
C1 = 100 pF.
V_{ZAP} = 400 V charge on C1.
S1 = Hg-wetted "bounceless" relay.

FIGURE 5. High voltage (V_{ZAP}) test circuits.

b. With S_1 in position 1, charge C_1 to V_{ZAP} and then switch to position 2. Apply the test voltage (V_{ZAP}) to the same terminal selected for leakage current measurements. Apply V_{ZAP} in a 3 pulse sequence as follows:

(1) Input (-) to V_{DD}

(2) Input (+) to V_{SS}

(3) Input (+) to Output

In each case, V_{ZAP} shall be applied by charging C_1 to V_{ZAP} with S_1 in position 1 and then switching to position 2.

c. Within 24 hours, repeat the I_{SS} measurement and the I_{IH} and the I_{IL} measurements on the same terminals as performed above. At this time a DUT exhibiting leakage currents in excess of the specified limits is defective.

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4.6 Data reporting. When specified in the acquisition document, a copy of the following data, as applicable, shall be supplied:

a. Attributes data for all screening tests (see 4.2) and variables data for all dynamic burn-in, and operating life tests (see 4.4.3).
b. The quality conformance inspection data (see 4.4).
c. Final electrical parameters data (see 4.2c).

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging of microcircuits shall be in accordance with MIL-M-38510.

6. NOTES

6.1 Notes. The notes specified in MIL-M-38510 are applicable to this specification.

6.2 Intended use. Microcircuits conforming to this specification are intended for logistic support of existing equipment.

6.3 Ordering data. The acquisition document should specify the following:

a. Complete part number (see 1.2).
b. Requirements for delivery of one copy of the quality conformance inspection data (see 4.6) pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.
c. Requirements for certificate of compliance, if applicable.
d. Requirements for notification of change of product or process to the contracting activity, if applicable.
e. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action and reporting results, if applicable.
f. Requirements for product assurance options.
g. Requirements for special carriers, lead lengths, or lead forming, if applicable. These requirements shall not affect the part number. Unless otherwise specified, these requirements shall not apply to direct purchase by or direct shipment to the Government.

6.4 Handling. CMOS microcircuits should be handled with certin precautions to avoid damage due to the accumulation of static charge. These CMOS devices are fabricated with a silicon gate technology, including input protection, which reduces the susceptibility to damage. However, the following handling practices are recommended:

a. Devices should be handled on benches with conductive and grounded surface.
b. Ground test equipment and tools.
c. Do not handle devices by the leads.
d. Store devices in conductive foams or carriers.
e. Avoid use of plastic, rubber, or silk in areas where CMOS devices are handled.
f. Maintain relative humidity above 50 percent, if practical.

6.5 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-STD-1313, MIL-STD-1331, and as follows:

V_{DD}	Supply voltage
V_{SS}	Common or reference voltage node
MA_0 through MA_8	Address input
$\overline{CS}$	Chip select input
BUS_0 through BUS_7	Data output

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NC	Not connected
GND	Ground. Zero voltage potential
T_A	Free air temperature
$V_{IC(pos)}$	Positive clamping input to V_{DD}
$V_{IC(neg)}$	Negative clamping input to V_{SS}
BKDN	Breakdown supply current
I_Z	Output high impedance leakage current
I_{SS}	Quiescent supply current
t_{AA}	Address access time
t_{AC}	Chip select delay, time from chip select to data valid
t_{CSZ}	Time from chip select to high impedance output

6.6 Logistic support. Lead materials and finishes (see 3.3) are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class B (see 1.2.2) and lead material and finish C (see 3.3). Longer length leads and lead forming shall not affect the part number.

6.7 Generic test data. Generic test data may be used to satisfy the requirements of 4.4.3. Group C generic test data shall be on date codes no more than one year old and on a die in the same microcircuit group (see appendix E of MIL-M-38510) with the same material, design and process and from the same plant as the die represented. Group D (see 4.4.4) generic data shall be on date codes no more than one year old and on the same package type (see terms, definitions, and symbols of MIL-M-38510) and from the same plant as the package represented. The vendor is required to retain the generic data for a period of not less than 36 months from the date of shipment.

6.8 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

Military device type	Generic-industry type
01	1832

6.9 Ordering guidance. Since the qualification and certification requirements have been removed from the specification, orders may be placed immediately.

6.10 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodian:
NASA - NA

Preparing activity:
NASA - NA

(Project 5962-K646-2)

Agent:
DLA - ES

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-38510/472A(NASA)	Microcircuits, Digital, CMOS, 512 Word x 8-Bit Static Read only Memory (ROM) Monolithic Silicon

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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